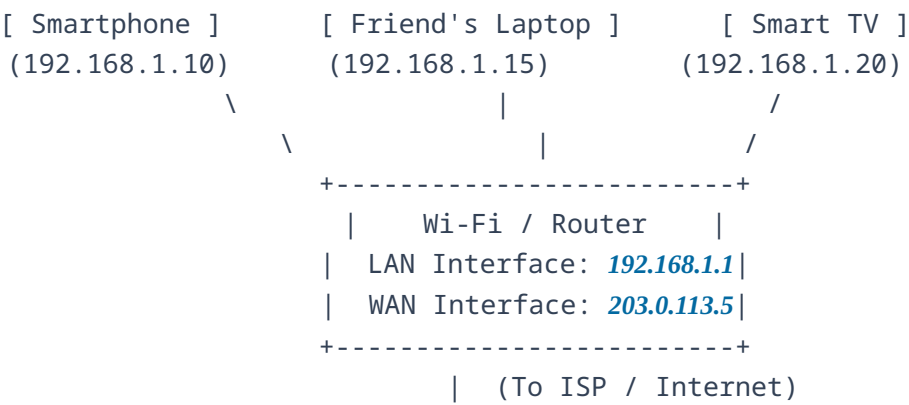


Computer Networks & Routing Analysis

Comprehensive Assignment: Network Topologies, Packet Routing, Routing Tables, Protocols, and Configuration

Task 1: Home Network Diagram & IP Addressing

In a standard local area network (LAN), devices connect to a central home router via Wi-Fi or Ethernet. The router acts as the default gateway, bridging the local subnet to the wider internet and managing local device communication.



Interface & Device IP Breakdown

Device / Interface	IP Address	Subnet Mask	Role
Smartphone	192.168.1.10	255.255.255.0	End User Device (Wi-Fi Client)
Friend's Laptop	192.168.1.15	255.255.255.0	End User Device (Wi-Fi Client)
Smart TV	192.168.1.20	255.255.255.0	IoT / Streaming Device (Ethernet/ Wi-Fi)
Router LAN Interface (Gateway)	192.168.1.1	255.255.255.0	Local Default Gateway for Subnet
Router WAN Interface	203.0.113.5	255.255.255.228	External Uplink to Internet Service Provider

Task 2: Step-by-Step Manual IP Routing Simulation

Consider a packet originating from **Laptop A** (Subnet 1: **192.168.1.50 / 24**) destined for a friend's computer **Laptop B** (Subnet 2: **10.0.5.20 / 24**) located in a different office building across a routed network.

Routing Journey & Step-by-Step Explanation

Step 1: Subnet Analysis at Source. Laptop A compares its own IP (**192.168.1.50**) with the destination IP (**10.0.5.20**) using its subnet mask (**255.255.255.0**). It performs a bitwise AND operation and determines that the destination is on a *different network*.

Step 2: Forwarding to Default Gateway. Because the destination is off-link, Laptop A cannot send an ARP request for Laptop B. Instead, it encapsulates the packet in an Ethernet frame addressed to the MAC address of its local router (**R1**, interface **192.168.1.1**).

Step 3: Router R1 Inspection & Routing Table Lookup. **R1** receives the packet, strips the layer 2 header, and examines the destination IP (**10.0.5.20**). It checks its routing table for the best match:

Destination	Subnet Mask	Next Hop	Interface
10.0.5.0	255.255.255.0	192.168.2.2 (R2)	Eth1 (WAN)

Step 4: Inter-Router Transit. **R1** forwards the packet across the point-to-point link to intermediate router **R2** (**192.168.2.2**).

Step 5: Final Delivery. Router **R2** checks its routing table, sees that **10.0.5.0/24** is directly connected to its **Eth2** interface, resolves Laptop B's MAC address via ARP, and delivers the packet directly to Laptop B (**10.0.5.20**).

Task 3: Zomato-Style Delivery Network Routing Table

A fictional multi-city food delivery platform operates distribution hubs in Mumbai, Bengaluru, and Delhi. Each branch operates on an isolated internal subnet, communicating via secure site-to-site VPN tunnels over routers.

Branch Location	Network Address	Subnet Mask	Next Hop	Interface
Mumbai Hub (HQ)	10.100.0.0	255.255.255.0	Directly Connected	Eth0 (Local LAN)
Bengaluru Branch	10.200.0.0	255.255.255.0	172.16.1.2 (ISP Gateway)	Tunnel0 (VPN WAN)
Delhi Branch	10.300.0.0 (10.3.0.0)	255.255.255.0	172.16.1.2 (ISP Gateway)	Tunnel0 (VPN WAN)

Default / Internet Route	0.0.0.0	0.0.0.0	172.16.1.1 (Core Gateway)	Eth1 (Internet Uplink)
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Task 4: Static Routing vs. Dynamic Routing Protocols

Routing strategies dictate how paths are maintained across networks. Below is a detailed comparison using real-world analogies such as food delivery routing engines (e.g., Zomato, DoorDash, Uber Eats).

Static Routing	Dynamic Routing (OSPF / RIP)
Pros 1. Zero CPU Overhead: Routers do not expend compute cycles running background discovery algorithms. 2. Predictable Paths: Traffic always follows explicit, human-defined paths without unexpected rerouting. 3. Enhanced Security: No routing updates are broadcasted, reducing the attack surface for route poisoning. Cons 1. Not Scalable: Adding new branch offices requires manual updates on every single router in the topology. 2. No Automatic Failover: If a primary link fails, traffic drops permanently unless a secondary static backup route is manually configured. 3. Rigid Architecture: Unsuitable for dynamic environments where routes change frequently.	Pros 1. Automatic Convergence: If a fiber line is cut, protocols instantly recalculate alternative healthy paths. 2. High Scalability: New nodes and subnets advertise themselves automatically without manual intervention. 3. Optimal Path Selection: Algorithms (like Dijkstra in OSPF) continuously evaluate metric costs to find the fastest path. Cons 1. Resource Intensive: Consumes router memory and CPU bandwidth for periodic hello packets and topology updates. 2. Misconfiguration Risks: A rogue router or bad metric advertisement can cause routing loops or black holes. 3. Harder Troubleshooting: Diagnosing transient routing flapping requires specialized packet capture and protocol logs.

Real-World App Example: Zomato / Food Delivery Dispatch A food delivery platform relies heavily on dynamic routing principles. If a main bridge or highway in a city becomes blocked due to an accident, static routing (fixed pre-planned delivery paths) would cause couriers to get stuck or fail deliveries. Dynamic routing protocols (like OSPF) mirror live delivery dispatch algorithms: when a link failure occurs, the routing engine instantly dynamically computes alternative roads, ensuring orders reach customers seamlessly.
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Task 5: OSPF Router Configuration & Annotation

Below is a standard OSPF (Open Shortest Path First) configuration between two routers, generated via AI and fully annotated for clarity.

```
! --- Router A Configuration ---
router ospf 1
  router-id 1.1.1.1
  network 192.168.1.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.3.255 area 0
  passive-interface GigabitEthernet0/0

! --- Router B Configuration ---
router ospf 2
  router-id 2.2.2.2
  network 192.168.2.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.3.255 area 0
  passive-interface GigabitEthernet0/0
```

Line-by-Line Configuration Explanation

Configuration Command	Detailed Explanation
<code>router ospf 1</code> / <code>router ospf 2</code>	Enables the OSPF routing process on the router and assigns a local process ID (1 or 2). This ID is locally significant to the router instance.
<code>router-id 1.1.1.1</code> / <code>2.2.2.2</code>	Uniquely identifies the router within the OSPF autonomous system. Used by the SPF algorithm as the stable anchor identifier.
<code>network 192.168.1.0 0.0.0.255 area 0</code>	Advertises local subnets matching the wildcard mask into OSPF Area 0 (the Backbone Area). Enables neighbor discovery on attached interfaces.
<code>passive-interface GigabitEthernet0/0</code>	Stops OSPF hello packets from being sent out of the specified user-facing LAN interface, preventing unnecessary traffic and improving security while still advertising the subnet.